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Lattice thermal conductivity and Young's modulus of XN_4 ($\text{X} = \text{Be}$, Mg and Pt) 2D materials using machine learning interatomic potentials

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The newly synthesized BeN_4 monolayer has introduced a novel group of 2D materials called nitrogen-rich 2D materials. In the present study, the anisotropic mechanical and thermal properties of three members of this group, BeN_4 , MgN_4 , and PtN_4 , are investigated. To this end, a machine learning-based interatomic potential (MLIP) is developed and utilized in classical molecular dynamics (MD) simulations. Mechanical properties are calculated by extracting the stress–strain curve and thermal properties by the non-equilibrium molecular dynamics (NEMD) method. The acquired results show the anisotropic Young's modulus and lattice thermal conductivity of these materials. Generally, the Young's modulus and thermal conductivity in the armchair direction are higher than in the zigzag direction. Also, the anisotropy of Young's modulus is almost constant at every temperature for BeN_4 and MgN_4 , while for PtN_4 , this parameter is decreased by increasing the temperature. The findings of this research are not only evidence of the application of machine learning in MD simulations, but also provide information on the basic anisotropic mechanical and thermal properties of these newly discovered 2D nanomaterials.

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1. Introduction

Following the fabrication of graphene as the first two-dimensional (2D) material in 2004,¹ developing novel materials of this kind and investigating their properties and behavior has attracted considerable attention from researchers in various fields of science, ranging from chemistry and physics to engineering. Today, a wide range of 2D materials (*e.g.* phosphorene,² borophene,³ h-BN,^{4–7} black phosphorene,^{8,9} and MoS_2 ,^{10,11}) with distinguished mechanical, thermal, electrical and optical properties have been introduced. In addition to the properties of these materials, their geometry also makes them promising structures with potential applications in nanoelectronics and optoelectronics.

Generally, experimental study is the main approach in investigating the behavior and properties of materials and structures, but when it comes to micro/nanoscales, due to their submicron

dimensions, using this approach encounters remarkable restrictions. So, theoretical and computer simulation-based methods in this regard have been applied rapidly, particularly after recent notable advances in computational tools. Molecular Dynamics (MD) simulation and density functional theory (DFT) are methods in this regard that have been successfully implemented in modeling various systems including vibration behavior^{12–17} mechanical and thermal properties^{18–27} and electronic properties.^{28–31} DFT is a time-consuming method providing the most accurate results, while MD simulation is less time-consuming but its results firmly rely on the interatomic potential functions. Machine-learning methods provide an opportunity to develop specific potential functions for each desired simulation leading to highly accurate results in the range of DFT results with less computational cost. Mortazavi *et al.*^{32–36} conducted various research studies on investigating the mechanical and thermal properties of micro/nano-scale materials through developing machine-learning interatomic potentials (MLIPs). Recently, Arabha *et al.*³⁷ published a comprehensive review paper on the application of MLIPs in the calculation of lattice thermal conductivity. In this study, the thermal conductivity of different 2D and 3D materials obtained from various approaches including experiments, DFT, MD, and MLIP has been compared and the significance of developing MLIP was indicated. Also, different machine-learning techniques in creating interatomic potentials and their characteristics were presented in

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that paper. Zuo *et al.*³⁸ in a comprehensive comparison study analyzed the efficiency and the computational costs of different MLIPs including MTP, SNAP, qSNAP, NNP, and GAP for Cu and Ni fcc metals, Li and Mo bcc metals, and Si and Ge semiconductors, which are representative of various material/structural properties. Accordingly, it can be said that all MLIPs in comparison to classical interatomic potentials provide the highest accuracy in estimating the energies/forces and properties. Botu *et al.*³⁹ presented a workflow and five main steps in generating MLIPs.

The interfacial thermal conductance across C_3N , C_3B , C_2N , C_3N_4 , and C_3N_4 carbon-based 2D structures was also investigated.⁴⁰ Mortazavi *et al.*⁴¹ proposed a new computational approach to compute a phonon dispersion relation and analyze the dynamical stability of nanomaterials. This approach is based on a machine-learning interatomic potential and provides faster and more efficient results than the DFT simulations. The usefulness of this approach has been presented for a wide range of low-symmetry and porous 2D nanomaterials.

Arabha and Rajabpour⁴² investigated the thermal conductivity and Young's modulus of nitrogenated holey graphene (C_2N) using MLIPs. The significant dependency of thermal conductivity on length was reported in their work. They have also evaluated the development of MLIPs for point-defected C_2N structures. They have shown that while there is uncertainty in the stability of point-defected C_2N structures using the Tersoff classical interatomic potential, the MLIP is able to well calculate the properties of these structures. It should be mentioned that machine learning methods have broad applications in the field of nanostructures in addition to developing interatomic potentials. For instance, Jin *et al.*⁴³ investigated the use of machine learning to optimize interfacial thermal transfer across the Si/Ge interface. Also, the machine learning methods for thermoelectric materials research were thoroughly reviewed by Wang *et al.*⁴⁴

Recently, the synthesis of BeN_4 ⁴⁵ introduced a new group of 2D materials called nitrogen-rich 2D materials with XN_4 chemical formulation, in which X is the representative of metallic elements.²⁸ Thereafter, Mortazavi *et al.*²⁸ analyzed the stability and the mechanical, thermal, electronic, and phononic properties of a wide range of these 2D materials including BeN_4 , MgN_4 , IrN_4 , RhN_4 , NiN_4 , CuN_4 , AuN_4 , PdN_4 , and PtN_4 by conducting DFT calculations. They have reported that among the observed materials, only BeN_4 , MgN_4 , IrN_4 , PtN_4 and RhN_4 have the required dynamical and thermal stability. Also, it has been shown that the mechanical properties of BeN_4 , MgN_4 , PtN_4 and RhN_4 are sensitive to the armchair and zigzag directions so that the armchair ones have a higher Young's modulus and tensile strength than those in zigzag structures. Berdiyorov *et al.*⁴⁶ investigated the anisotropic electronic properties of BeN_4 and MgN_4 using the DFT and non-equilibrium Green's functional methods. They have indicated that in comparison to the zigzag direction, the armchair direction provides remarkably larger electronic charge transport capability. Cheng *et al.* investigated the lattice thermal conductivity of layered Dirac semimetal BeN_4 using the Boltzmann transport equation and first-principles three-phonon calculations.⁴⁷ Wang *et al.*⁴⁸

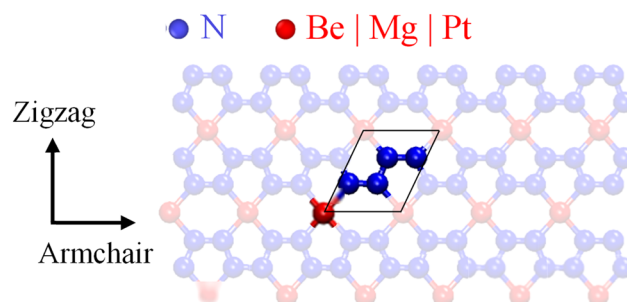


Fig. 1 Atomic structure of BeN_4 , MgN_4 , and PtN_4 2D monolayers and their corresponding armchair and zigzag directions.

used first-principles simulations to investigate the thermal properties of BeN_4 and MgN_4 . They demonstrated that the lattice thermal conductivities of BeN_4 and MgN_4 monolayers are anisotropic at 300 K, and the BeN_4 monolayer has a higher lattice thermal conductivity than the MgN_4 monolayer.

In the present study, a machine learning-based interatomic potential is developed and employed in MD simulations for investigating the mechanical properties and thermal conductivity of BeN_4 , MgN_4 , and PtN_4 2D monolayers as shown in Fig. 1. The main steps in this regard are as follows. Firstly, the *ab initio* molecular dynamic trajectories are obtained as the subsample set, and then, the interatomic potential is trained over this subsample using the momentum tensor potential (MTP) machine learning method. Finally, utilizing the trained interatomic potential in the MD simulation, the desired mechanical and thermal properties are calculated.

2. Methodology

In this section, the process of extracting the machine learning-based interatomic potentials using the *ab initio* molecular dynamic (AIMD) results and MTP method proposed by Shapeev,⁴⁹ and utilizing them in classical MD simulations to calculate the mechanical properties and thermal conductivity of BeN_4 , MgN_4 , and PtN_4 will be discussed.

2.1. *Ab initio* simulation

In this work, the GGA/PBE functional⁵⁰ and Vienna *Ab initio* Simulation Package (VASP) were utilized for AIMD simulation. Structures are firstly optimized for seven different strains between 0–14% and ten various temperatures between 100–1000 K. We considered plane-wave cutoff energies of 500 eV for the considered structures. It should be noted that all simulations were performed 1000 times with steps of 1 fs, at every strain and temperature for $5 \times 4 \times 1$ supercells with 125 atoms.

2.2. Interatomic potential training

Here, the MLIPs are developed using the MTP.⁴⁹ According to this method, the interatomic potentials can be defined as a multiplication of inertia tensors of radial polynomial functions. In order to train the parameters of MTP, the difference between the

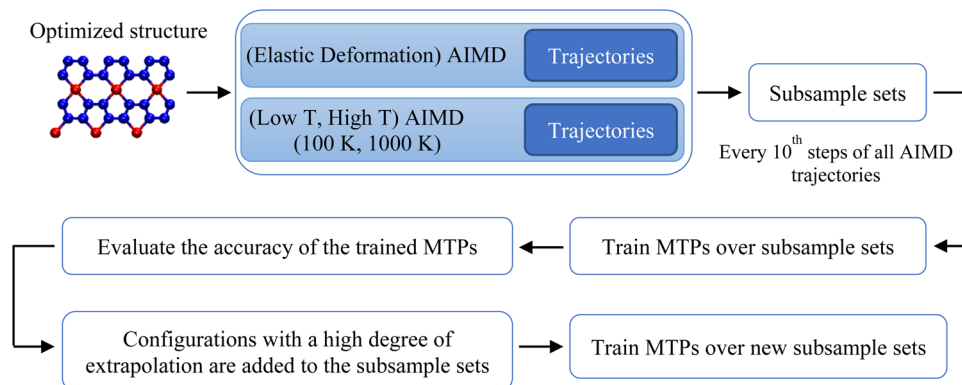


Fig. 2 Machine-learning interatomic potentials (MLIPs) training algorithm.⁴²

obtained results from the first principle calculations (*i.e.* energy, forces, and stresses) and the predicted results is minimized:^{49,51}

$$\sum_{m=1}^M \left[w_e (E_m^{\text{AIMD}} - E_m^{\text{MTP}})^2 + w_f \sum_i^N |f_{m,i}^{\text{AIMD}} - f_{m,i}^{\text{MTP}}|^2 + w_s \sum_{i,j}^N |\sigma_{m,ij}^{\text{AIMD}} - \sigma_{m,ij}^{\text{MTP}}|^2 \right] \rightarrow \min \quad (1)$$

In this equation, m is the number of training set configurations and N is the total number of atoms. E_m , $f_{m,i}$, and $\sigma_{m,i}$ are respectively energy, force, and stress terms, and AIMD and MTP superscripts indicate the corresponding values obtained from AIMD modeling and MTP calculations. Also, w_e , w_f , and w_s are the positive weighted coefficients showing the significance of energy, force, and stress expressions, and are considered to be 1, 0.1, and 0.001, respectively.

For the initial training subsample sets, 10% of the AIMD calculated trajectories at first are considered in this regard. This is due to the fact that the correlation of the calculated trajectories is done in a short-range time. Then, MTP is trained over these initial subsample sets. By assessing the accuracy of the trained MTP over all of the AIMD trajectories, the subsample sets are updated by the high degree of extrapolation trajectories,⁵² and again MTP is trained over these new subsample sets. The algorithm of training the MLIPs is indicated in Fig. 2. The ultimate trained MTP is considered as the MLIP and used in MD simulations.

2.3. Molecular dynamic simulations

In this paper, MD simulations are performed using a Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) package,⁵³ and the MTP trained potential as the interatomic potential, with a 1 fs time-step and 300 K average temperature. Also, the boundary conditions are periodic, which removes the surface effects at the boundaries and minimizes the influence of finite-length.

The lattice thermal conductivity of the desired materials is calculated based on the non-equilibrium molecular dynamics (NEMD)^{54–57} method. To this end, at first by the use of a Nose–Hoover barostat and thermostat at room temperature, the nanostructures are relaxed for 2.5 ps. Then, the heat flux is imposed in the equilibrated structures by creating a

temperature gradient between the defined hot and cold regions at the end sides of the structures, as shown in Fig. 3. The added

and subtracted average energy from the two baths are calculated in the microcanonical ensemble (NVE) for 2 ns. At the end, the thermal conductivity is calculated using Fourier's law of heat conduction as follows:

$$\kappa = \frac{-q''}{\frac{dT}{dx}} \quad (2)$$

In which q'' is the heat flux through the structure and $\frac{dT}{dx}$ is the temperature gradient within it.

In order to compute the mechanical properties, the simulation box at first is stretched in the direction of loading at a specific engineering strain rate of $1 \times 10^9 \text{ S}^{-1}$, and then by plotting the stress–strain curve, the slope in the linear region is reported as the Young's modulus of the structure. Based on the virial theorem, the stress values are obtained as follows:⁵⁸

$$S = \frac{1}{V} \sum_{a \in V} \left[-m \vec{v}_a \otimes \vec{v}_a + \frac{1}{2} \sum_{a \neq b} \left(\vec{r}_{ab} \otimes \vec{F}_{ab} \right) \right] \quad (3)$$

in which, S represents the stress tensor and V is the volume of the structure. m and $\vec{v}_a$ are the mass and velocity vector, respectively. Also, $\vec{r}_{ab}$ and $\vec{F}_{ab}$ indicate the position and force vectors between atoms a and b , and $\otimes$ denotes the outer product.

The cumulative energy variations of the hot and cold domains are respectively shown in Fig. 4 as a function of time. The system approaches a stationary state because the total amount of energy exchanged between hot and cold regions must eventually be zero over time. Also, Fig. 4 depicts the temperature profile inside the structure. A linear temperature gradient is generated along the structure by neglecting non-linear temperature jumps near the hot and cold regions.

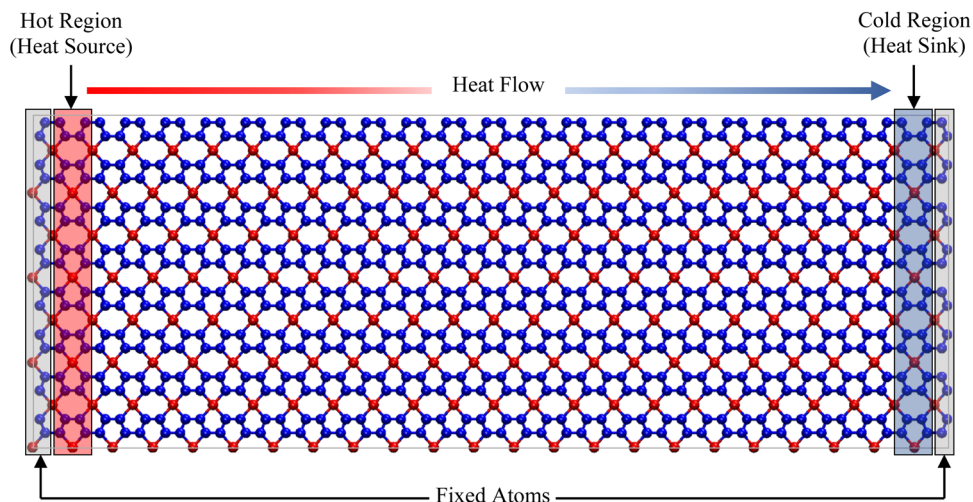


Fig. 3 Schematic illustration of the NEMD setup for the calculation of thermal conductivity.

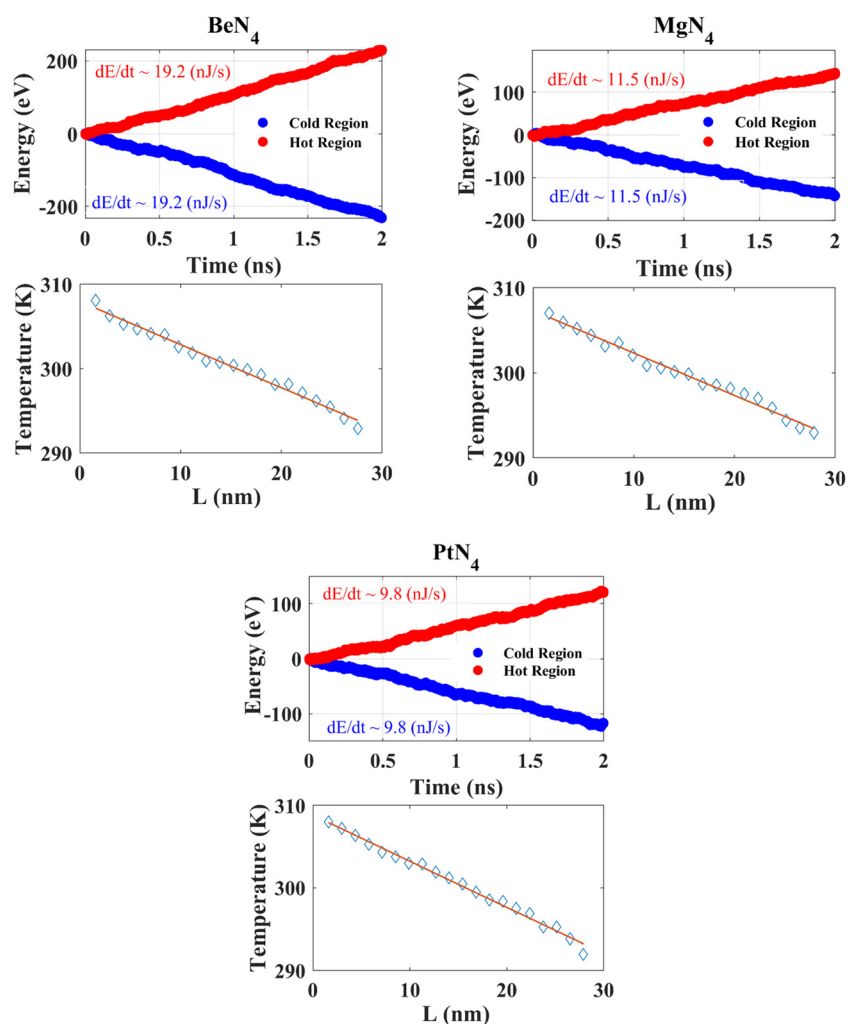


Fig. 4 Accumulative energy variations and temperature profile of BeN₄, MgN₄, and PtN₄ (zigzag direction) after the transient period.

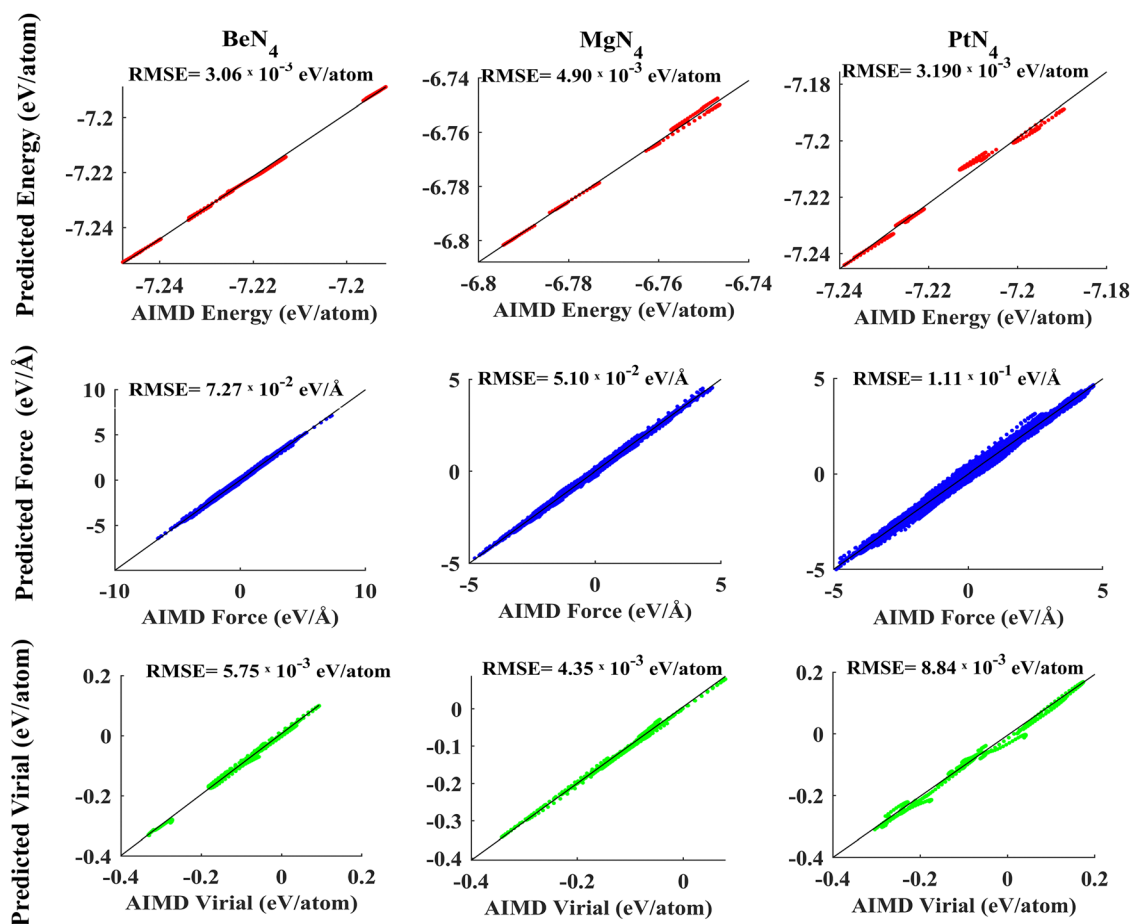


Fig. 5 Comparisons of AIMD and MLIP predicted energy, forces, and virial stress on configurations selected from the test data sets.

3. Results and discussion

The main focus of the present research is the calculation of the thermal and mechanical properties of BeN₄, MgN₄, and PtN₄ monolayers using MD simulation and MLIP. The acceptable data set for training the MLIP must consist in all the possible trajectories that a system may experience in MD simulation. So, to generate training data sets, a wide range of temperatures, from 100 K to 1000 K, are considered for the AIMD simulations. This range of temperature provides trajectories with both long-wavelength phonons, related to low-temperature conditions, and high-frequency optical modes, related to high-temperature conditions. Also, the high-temperature AIMD trajectories are needed for the consideration of large local deformations that probably happen in NEMD simulations of long-length structures.

After the training process, the testing data (200 configurations) were used to verify model accuracy. Fig. 5 depicts the relationship between per-atom energy, force, and per-atom virial stress that AIMD and MTP predicted. Root-mean-squared errors (RMSE) are quite low for energy, force, and virial stress (with a correlation coefficient of 0.99), indicating that the trained MTP has similar accuracy to AIMD.

3.1. Thermal properties

By implementing the trained MLIP in NEMD simulations, the lattice thermal conductivity of BeN₄, MgN₄, and PtN₄ 2D nanosheets in the armchair and zigzag directions has been investigated. The thermal conductivity of 1D materials, such as carbon nanotubes and semiconductor nanowires, as well as 2D nanomaterials (*e.g.*, graphene), has been found to be significantly size-dependent.⁵⁹ In Fig. 6, the variation of thermal conductivity *versus* length in both the armchair and zigzag directions is shown. As can be seen in this figure, for all these three materials and in both directions, the thermal conductivity at short lengths is length-dependent so that by increasing the length, the thermal conductivity increases. But, as the length increases, this dependency decreases and thermal conductivity converges to a specific value at longer lengths. The values of thermal conductivity in both directions are presented in the same figure.

From a comparative point of view, it can be seen that the BeN₄ monolayer has the highest thermal conductivity in both directions with ~ 140 and ~ 107 W m⁻¹ K⁻¹ in the armchair and zigzag directions, respectively, while the lowest values of this property in the armchair and zigzag directions belong to the PtN₄ monolayers with ~ 115 and ~ 41 W m⁻¹ K⁻¹, respectively.

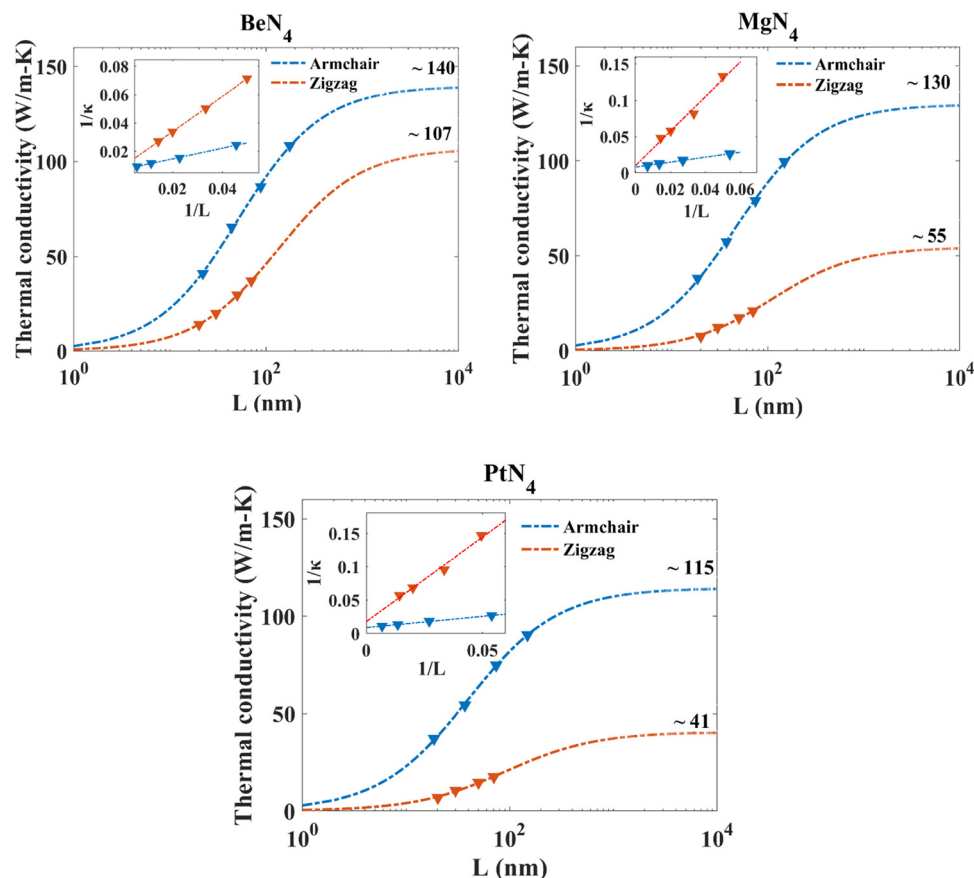


Fig. 6 The variation of lattice thermal conductivity as a function of length at 300 K.

Also, the change rate of thermal conductivity relative to length is different in the armchair and zigzag directions and there is a significant difference in the values of thermal conductivity between these two directions. In other words, it can be stated that in thermal applications, the orientation of these 2D materials is of great importance. It should be noted that the thermal conductivity at infinite length, k_{∞} , is approximated through fitting of the well-known equation $1/k_L = (1 + L/L)/k_{\infty}$ ⁶⁰ on the calculated thermal conductivity at finite lengths, k_L . In this equation, L is the effective phonon mean free path.

The observed thermal behavior is justifiable by analyzing the phonon group velocity as the main criterion of lattice thermal conductivity variation. To this end, the phonon group velocities of BeN₄, MgN₄, and PtN₄ are plotted in Fig. 7. Also, in this figure the phonon group velocity of graphene as a benchmark is presented. The phonon group velocities at low-frequency modes are generally considered as the main heat carrier. As it is obvious in this figure, at low-frequency modes, BeN₄ and PtN₄, respectively, have the highest and lowest phonon group velocities, which is indicative of their thermal conductivity. This result is completely aligned with the obtained lattice thermal conductivity.

3.2. Mechanical properties

In this subsection, temperature and chirality-dependent mechanical properties of BeN₄, MgN₄, and PtN₄ 2D materials

obtained from classical MD simulations with the trained interatomic potential will be discussed. To this end, the uniaxial stress-strain in the armchair and zigzag directions with a size of 5 nm × 5 nm for these materials (see Fig. 1) at room temperature is evaluated and shown in Fig. 8. It should be noted that the thickness of structures is assumed to be 3.06 Å. According to this figure, it can be said that for these 2D materials, the ultimate strength in the zigzag direction is generally lower than those of the armchair ones. Also, as can be concluded from this figure, the ultimate strengths of BeN₄, MgN₄, and PtN₄ in the armchair direction are approximately close together (with about 70, 76, and 76 GPa, respectively). The same conclusion is not valid in the zigzag direction, where just the ultimate strength of BeN₄ and MgN₄ ($\approx$ 40 GPa) are similar in this direction while for PtN₄ the value of this property is about 70 GPa. The reason for such a behavior and generally for the anisotropic mechanical behavior of these structures is rooted in the dominant bonds that have to be broken during the uniaxial loading. As it is clear in Fig. 1, the N–N bond is perfectly aligned in the armchair direction so in the case of uniaxial loading along this direction, it can be said that initiation of the bond failure as the criterion in calculating the ultimate strength depends mostly on the N–N bonds regardless of the type of metallic atoms (Be, Mg, and Pt) in the structure. On the other hand, in the zigzag direction, the Be–N, Mg–N,

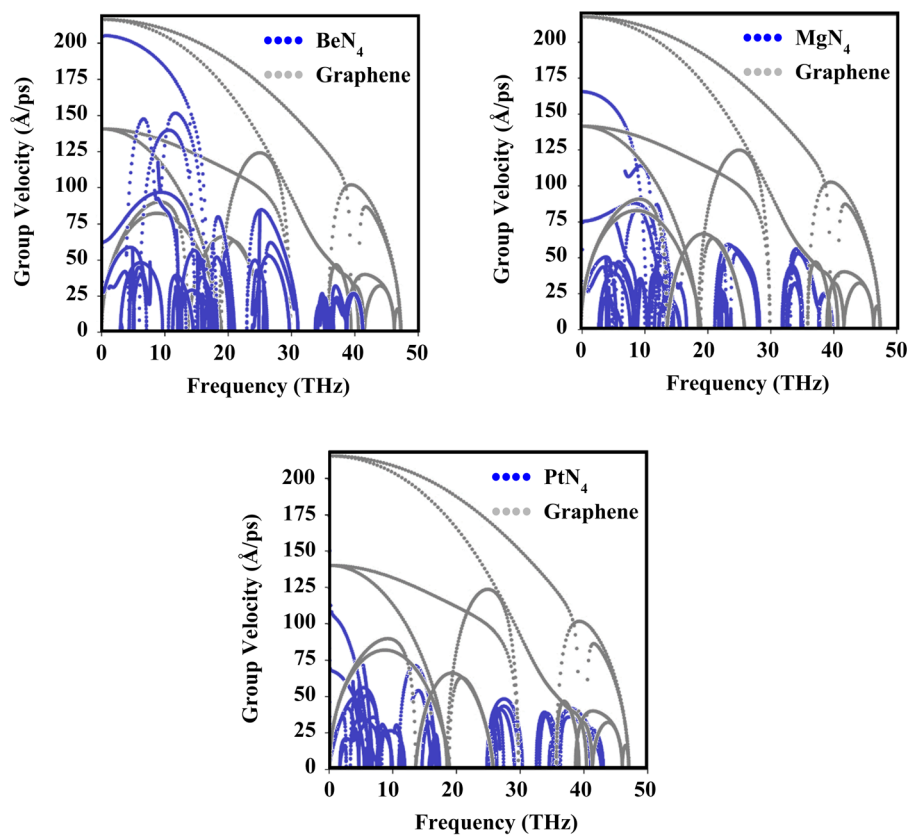


Fig. 7 Phonon group velocity. Comparison of BeN₄, MgN₄, and PtN₄ with graphene.

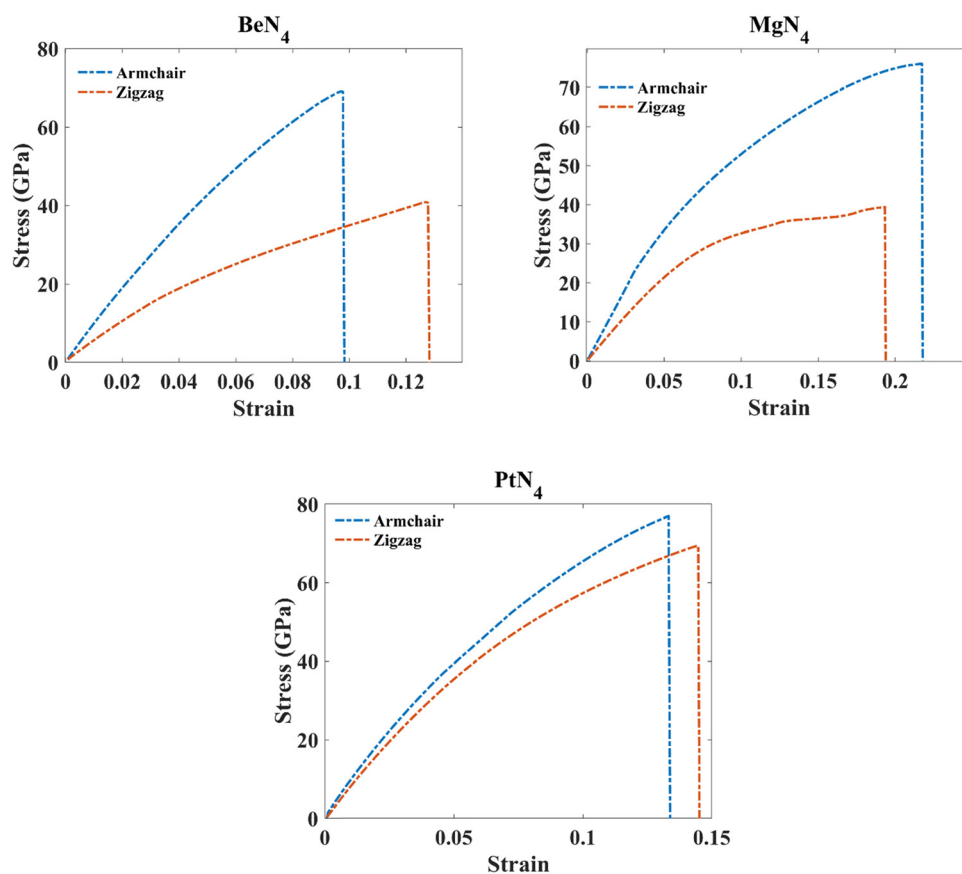


Fig. 8 Stress-strain diagram under uniaxial tensile load at 300 K.

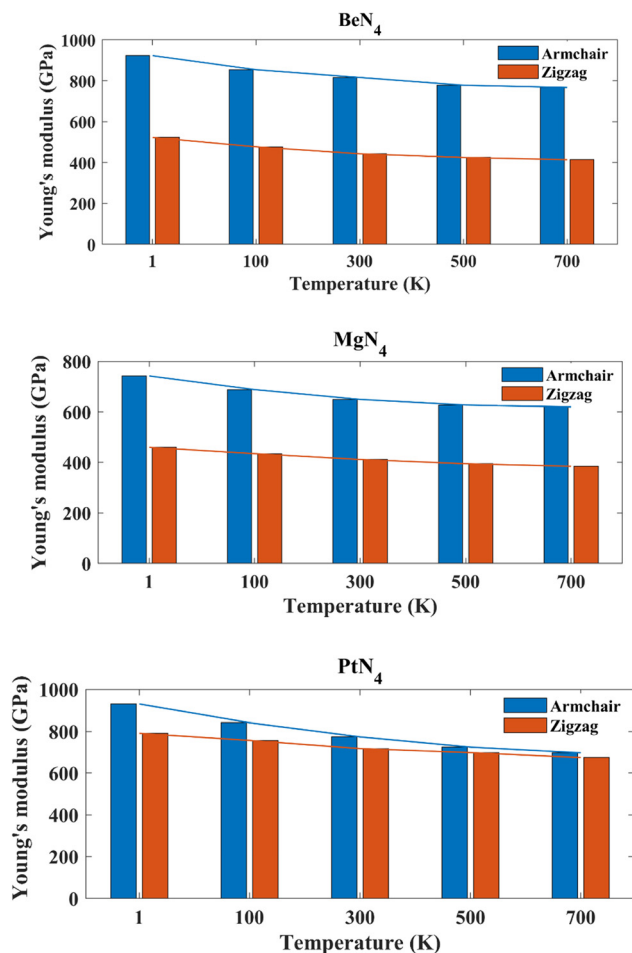


Fig. 9 The variation of Young's modulus as a function of temperature.

and Pt–N bonds play a dominant role so that the type of metallic atoms has an influence on the ultimate strength.

In contrast to MgN_4 , the fracture strain in the zigzag direction is higher than in their armchair counterparts. Also, it can be deduced from this figure that MgN_4 and BeN_4 have the highest and lowest values of fracture strain, respectively.

In Fig. 9, the variation of Young's modulus *versus* temperature has been presented. The dependency of Young's modulus on chirality and temperature is obvious in this figure. As illustrated in this figure, armchair structures have higher values of Young's modulus than those of the zigzag structures, and by increasing the temperature, the Young's modulus decreases in both the armchair and zigzag directions. Moreover, it can be noticed that both the armchair and zigzag structures of MgN_4 have the lowest values of Young's modulus, while interestingly, the highest values of this property are associated with BeN_4 in the armchair direction and PtN_4 in the zigzag direction. To have a better illustration of the Young's modulus of these materials and comparison between them, the Young's modulus of BeN_4 , MgN_4 , and PtN_4 along the armchair and zigzag directions at 300 K is also presented in Fig. 10.

In order to examine the dependency of the anisotropic behavior on temperature, the variation of anisotropy relative

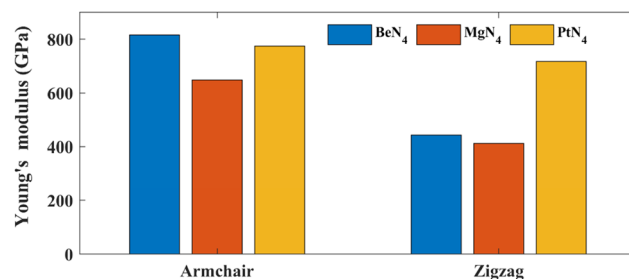


Fig. 10 Comparison of Young's modulus of BeN_4 , MgN_4 , and PtN_4 in the armchair and zigzag directions at 300 K.

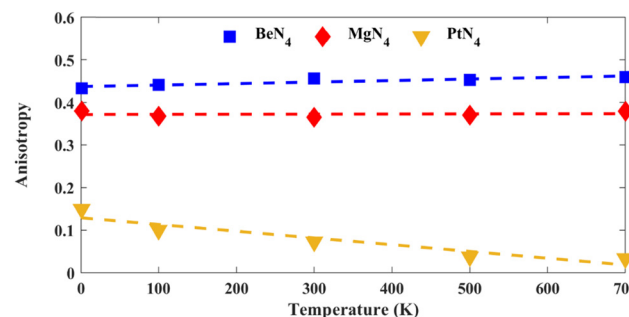


Fig. 11 Dependency of anisotropy on temperature.

to temperature is plotted in Fig. 11. Here, anisotropy is defined as $\frac{E_a - E_z}{E_a}$ where E_a and E_z are the Young's modulus along the armchair and zigzag directions, respectively. As can be seen in this figure, for BeN_4 and MgN_4 , the anisotropy is about 0.4 and almost constant at any temperature, while for PtN_4 , the anisotropy is less than 0.2 and decreases by increasing the temperature. This behavior is also obvious in Fig. 9 more specifically for PtN_4 . So, it can be stated that BeN_4 and MgN_4 display an anisotropic mechanical behavior at each temperature, but PtN_4 shows an isotropic mechanical behavior at high temperatures.

4. Conclusion

In summary, the present study provides details on the anisotropic mechanical and thermal properties of BeN_4 , MgN_4 , and PtN_4 monolayers, a new group of 2D materials. This aim was achieved by developing a MLIP using the MTP method, and MD simulation. The anisotropic Young's modulus and lattice thermal conductivity are concluded from the results in such a way that armchair orientation leads to higher Young's modulus and thermal conductivity than the ones in the zigzag orientation.

Also, the anisotropy $\left(\frac{E_a - E_z}{E_a}\right)$ in BeN_4 and MgN_4 is independent of the temperature and is approximately constant. However, increasing the temperature reduces anisotropy in PtN_4 . Moreover, thermal conductivity increases by increasing the length and at long lengths, it reaches a constant value so that BeN_4 presents the highest thermal conductivity in the armchair and zigzag directions, and PtN_4 exhibits the lowest values in both directions.

On the other hand, the order from highest to lowest values of Young's modulus follows BeN₄, PtN₄, and MgN₄ in the armchair direction and PtN₄, BeN₄, and MgN₄ in the zigzag direction, while the trend for thermal conductivity is BeN₄, MgN₄, and PtN₄ in both directions.

Furthermore, the present study is evidence of the application of the machine learning approach in developing interatomic potential functions and their use in classical MD simulations yielding the highest accuracy with less computational costs.

Conflicts of interest

There are no conflicts to declare.

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